HW 2

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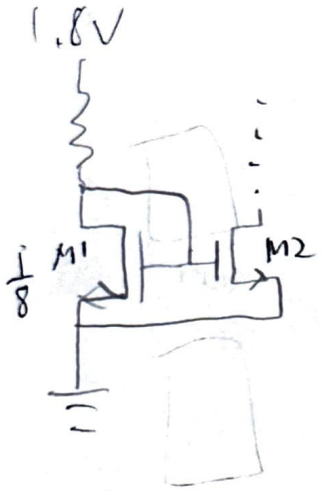
E/EN 252

1.



$K' = 160\ \mu A/V^2 \quad \lambda = .01 \quad V_T = .6V$

$6.4 \times 10^{-6} = \frac{1}{2} 160 \times 10^{-6} \cdot \frac{1}{8} \cdot V_{ov}^2$

$V_{ov} = .8V$

$.8 + .6 = 1.4V$ at D and G

a. $\frac{1.8 - 1.4}{R_{bias}} = 6.4 \times 10^{-6}$

$\boxed{R_{bias} = 62500\Omega}$

b. $\frac{32}{6.4} = 5$

$\left(\frac{1}{8}\right)5 = \frac{5}{8}$

$W = 5\mu m$

c. min is V_{ov}

$\boxed{.8V}$

d. $r_{o2} = \frac{1}{\lambda I_D} = \frac{1}{.01(32 \times 10^{-6})} = 3.125 M\Omega$

e. $V_{GT} = .8V \quad V_{DS} = 1V$

so still in sat

therefore

I_D still $\approx 32 \times 10^{-6} A$

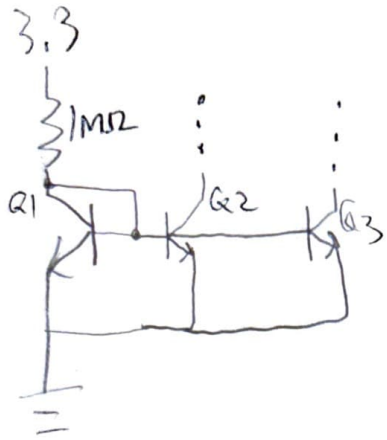
but technically affected

by channel length mod

so actually $32 \times 10^{-6}(1 + .01(1)) =$

$\boxed{32.32 \times 10^{-6} A}$

2.



$B = 100 \quad V_{be} = .7V$

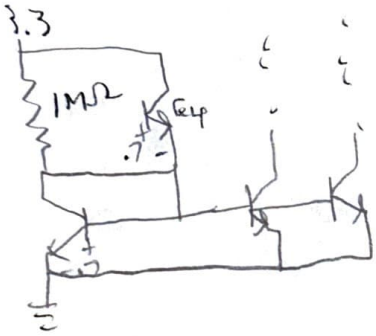
$$\frac{3.3 - .7.6}{1M\Omega} = 2.6 \times 10^{-6} A$$

$$\frac{I_{corr}}{I_{Bias}} = \frac{1}{1+\frac{N}{B}} = \frac{1}{1+\frac{3}{100}} = .9709$$

$.9709(2.6) = 2.524$

a. I_C in Q_2 and Q_3 = $2.524 \times 10^{-6} A$

b.

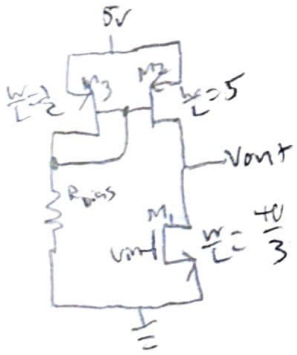


$3.3 - .7 - .7 = 1.9$

$$\frac{1.9}{1M\Omega} = 1.9 \times 10^{-6} A$$

c. $\frac{1.9}{R_{bias}} = 2.6 \times 10^{-6} \rightarrow R_{bias} = 730769.231\Omega$

3.



$k_n' = 150 mA/V^2 \quad V_{Tn} = V_{Tp} = .6V \quad \lambda_p = .01$

$k_p' = 100 mA/V^2 \quad \lambda_n = .02$

a. $g_m = k'_n \frac{W}{L} V_{ov} \Rightarrow 800 \times 10^{-6} = 150 \times 10^{-6} \left(\frac{40}{3}\right) V_{ov}$

$.4V = V_{ov}$

$V_{in} = V_{TN} + V_{ov} = .4 + .6 = 1V$

$I_{D1} = \frac{1}{2}(150 \times 10^{-6}(\frac{40}{3}) .4^2) = 1.6 \times 10^{-4}$

$I_{D1} = I_{D2} = 10 I_{D3}$

$I_{D3} = 1.6 \times 10^{-5}$

$1.6 \times 10^{-5} = \frac{1}{2}(100 \times 10^{-6}(\frac{1}{2}))V_{ov}^2$

$V_{ov} = .8$

$.8 + .6 = V_{gs} = 1.4$

$$\frac{(5 - 1.4) - 0}{R_{bias}} = 1.6 \times 10^{-5}$$

$R_{bias} = 225 k\Omega$

b.

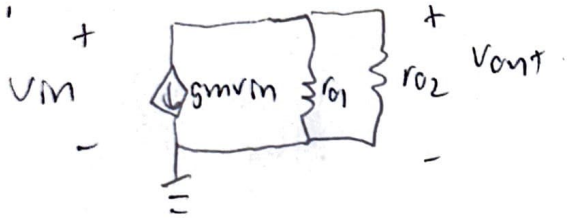
$V_{ov1} = .4$

$V_{ov3} = V_{ov2} = .8$

$.4 - 0$ to $5 - .8$

$.4 < V_{out} < 4.2$

c.



d. $A_v = -800 \times 10^{-6} (r_{o1} \| r_{o2})$

$r_{o1} = \frac{1}{.01(1.6 \times 10^{-4})} = 625 k\Omega$

$r_{o2} = \frac{1}{.02(1.6 \times 10^{-4})} = 312.5 k\Omega$

$A_v = -166.\bar{6}$ V/V